

How AI Alleviates Woes: The Impact of Chatbot Relational Cues on Emotional Interaction Experiences

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Conflict of Interest

The authors are unaware of any conflicts of interest, financial or otherwise.

Author contributions

YX, TG and YZ conceived the research, designed the experiment. YX and ZC analyzed the data. YX and ZC performed the experiment. YX, TG, YZ, ZC and RG wrote the manuscript.

Ethical standards

All procedures performed in this study were in accordance with the 1964 Helsinki declaration and its later amendments or comparable ethical standards. The local ethics committee approved the experimental protocol.

Consent to participate

Informed consent was obtained from all participants included in the study. All data generated or analyzed during this study are included in this published article.

Consent to publication

All participants and authors of the present study gave their consent for publication of this paper.

Data availability

The datasets generated during the current study are available from the corresponding author, Dr. Tian Gan (email: gantian@zstu.edu.cn) on reasonable request.

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Abstract

The relationship between humans and AI has evolved into a phase of deep integration, with the latest generation of text-based chatbots increasingly serving as trusted companions for emotional support. Relational cues (features in chatbots designed to simulate human emotional and social behaviors) reflect the core characteristics of human social existence and play a crucial role in enhancing human-AI emotional interactions. This research employs a mixed-methods approach to explore how relational cues shape human-AI emotional interaction experiences. Study 1 selected common indicators of emotional interaction experience and used an experimental method ($n = 196$), which showed that relational cues significantly increased users' emotional arousal, system usability, emotional attachment and human-AI trust. Moreover, mind perception exerted a negative moderating effect on the relationship between relational cues and pleasure, arousal, dominance, emotional attachment and usability. Furthermore, the efficacy of specific types of relational cues (particularly empathy) was stronger in extreme negative emotional events compared to moderate ones. Study 2 used an online survey method ($n = 25$) and further revealed that relational cues can improve mood, enhance perceived realism, foster intimacy, and elevate conversational quality. These improvements collectively contributed to enhancements in multifaceted emotional interaction experiences. It is worth noting that empathy cues exhibited the most positive effect in both studies. These findings highlight the beneficial role of relational cues in human-AI emotional interactions and their boundary conditions, providing a theoretical foundation for emotional agent design from the perspective of chatbot, users, and event.

Key words chatbot, anthropomorphism, relational cues, emotional interaction, mind perception

1. Introduction

Human emotional interaction has long been a cornerstone of social relationships, playing a crucial role in various aspects of life, from personal well-being to social culture (Burkitt, 1997). However, as society has evolved, factors such as the fast-paced lifestyle, population aging, and migration have made it increasingly difficult to fulfill people's emotional interaction needs, potentially leading to the rising prevalence of mental health problems (Kirkbride et al., 2024). The emergence of large language models (LLM) has accelerated the development and popularity of text-based artificial intelligence (AI) chatbots (Bhayana, 2024), which are expected to fulfill human emotional interaction needs. These models, with performance superior to most humans and continually improving emotional intelligence (Wang et al., 2023), show significant potential in facilitating AI chatbot's emotional interaction capabilities, driving the generalization of emotional interactions from person-to-person to person-to-AI (Shank et al., 2025). Given these advancements, improving the interaction experiences to provide timely and effective psychological support for individuals in need has emerged as a critical issue requiring immediate attention.

Building on the Hedonic-Pragmatic model (Hassenzahl, 2007), the human-AI emotional interaction experience refers to individuals' subjective perception of their emotional distress being recognized, understood, and appropriately addressed during interactions with AI chatbots. Previous research has shown that the interaction experiences between humans and chatbots are influenced by factors such as the human cues, mind perception, and conversation event characteristics (Kowatsch et al., 2018; Lee & Hahn, 2024; Shams & Kim, 2024). However, most prior studies were based on earlier rule-based chatbots or even relied on participants imagining interaction experience through

hypothetical conversation scenarios. This approach is increasingly limited, particularly in light of evidence showing that certain factors affect user interaction experience differently between rule-based and LLM-based AI chatbots (Pataranutaporn et al., 2023). Indeed, LLM-based AI exhibits unprecedented capabilities in language understanding, logical reasoning, and content generation, making it easier to foster complex, emotionally engaged, and trusting relationships with users (Huang et al., 2025; Lee et al., 2024) while becoming more effective in improving mental health (H. Li et al., 2023). Despite this substantial advancement in capabilities, there remains a notable gap in research examining how these factors influence the emotional interaction experiences between humans and LLM-based AI. To address this gap, the present research aims to explore the relationship between these factors and the multidimensional indicators of human-AI emotional interaction experience, with the goal of enhancing AI's emotional support capabilities.

1.1. Relational cues and human-AI emotional interaction experiences

Human cues of chatbots refer to features resulting from their simulation of human appearance, speech, and behavior, including visual cues (such as appearance, gender, age), verbal cues (such as text, voice, tone and speed), quasi-nonverbal cues (such as emoticons), and relational cues (Kowatsch et al., 2018). Previous research has extensively examined the impact of the first three types of cues on interaction experience. For example, visual cues have been shown to enhance users' perception of social presence, trust, and satisfaction with chatbots (Chen et al., 2024). Verbal cues increase social presence and emotional connection (Araujo, 2018). Additionally, quasi-nonverbal cues like emoticons enhance satisfaction by increasing the perception of warmth (Yu & Zhao, 2024). In contrast, existing research has not yet systematically explored the impact of relational cues on

emotional interaction experiences. Unlike the first three types of cues, which emphasize surface-level features of interaction, relational cues reflect the core characteristics of human social existence and play a crucial role in enhancing digital emotional support or human-AI relationship (Bradley & Lang, 1994; Mo et al., 2023; Skjuve et al., 2022). This kind of cue focuses on mimicking emotions and social behaviors inherent in human social interaction, and consists of five types: self-disclosure, empathy, humor, meta-talk, and continuity of conversation (Kowatsch et al., 2018; Mariacher et al., 2021). As human-AI relationships evolve toward deeper integration, the role of relational cues becomes increasingly important (Weng et al., 2025).

Many studies grounded in social presence theory and Computers are Social Actors (CASA) paradigm argue that relational cues have positive effect on human-AI interaction. When chatbots display specific cues (such as emotional signals), individuals tend to attribute a social actor identity to the chatbot, similar to that of a human (Nass et al., 1994; Wang et al., 2024), thereby increasing the sense of coexistence, human-AI intimacy (Weng et al., 2025), reducing psychological resistance (Mou et al., 2019), and increasing the likelihood of use (S. Lee et al., 2020). Existing research has validated the positive impact of specific types of relational cues on user interaction experience. For example, self-disclosure encourages deeper user self-disclosure, enhancing intimacy and pleasure during interactions (Y. C. Lee et al., 2020). Chatbots with high empathy can more effectively recognize and respond to users' emotions, thereby enhancing pleasure, arousal, and interaction satisfaction (Inzlicht et al., 2024; Liao & Yan, 2022; Liu & Sundar, 2018; Liu et al., 2024). Humor, particularly in greetings, strengthens emotional attachment and boosts engagement (Lekaviciute et al., 2023; Warren et al., 2018). Additionally, meta-talk contributes to a more engaging experience

by clarifying misunderstandings or managing conversational flow (Schuetzler et al., 2020). Lastly, continuity maintains coherence across multiple turns, improving naturalness and human-like perception (Go & Sundar, 2019).

However, some recent studies highlight the potential negative effect of relational cues in chatbots. Specifically, chatbot self-disclosure can diminish trust and fail to alleviate individuals' stress when emotional support is lacking, as individuals may feel that their stress is not being acknowledged in such situations (Meng & Dai, 2021). Furthermore, empathetic and sympathetic responses from chatbots may have a backfiring effect, reducing the chatbot's perceived authenticity, which results in detrimental effects on the willingness to trust and the intention to use it (Seitz, 2024). To address these heterogeneous findings, this research focuses on the exact role of relational cues, raising the first question: *Do high versus low levels of relational cues exhibited by AI chatbots enhance or diminish emotional interaction experiences? (Q1)*

1.2. The role of mind perception

In addition to the relational cues of chatbots, individuals' mind perception of chatbots is also a critical factor influencing interaction experience (Lee & Hahn, 2024). Mind perception refers to individuals' perceptions of psychological abilities in various entities (such as the ability to perceive emotions or the capacity to act based on subjective intentions, i.e., experience and agency) (Gray et al., 2012). This perception of psychological abilities in AI chatbots directly impacts the interaction process (S. Lee et al., 2020; Yam et al., 2021). For instance, participants who perceived a caring motive for the AI also rated it as more trustworthy, empathetic, and better-performing (Pataranutaporn et al., 2023). Besides, mind perception may play a moderating role between

relational cues and emotional interaction experiences. For example, among participants who were skeptical about the emotional abilities of chatbots, high levels of empathy cues displayed by chatbots significantly enhanced emotional support effects; however, for participants who trusted the chatbot's emotional abilities, this effect was not observed (Liu & Sundar, 2018). Moreover, research suggests that individuals who perceive AI as having strong psychological abilities are more likely to attribute "humanness" to it, reducing the impact of actual human traits (such as relational cues in this research) on interaction experience (S. Li et al., 2023). Additionally, research also reveals that individuals who believe chatbots possess psychological abilities are more likely to perceive a threat to the uniqueness of their own mind; when chatbots display relational cues, it can easily trigger a strong sense of eeriness or the uncanny valley effect, leading to a negative emotional interaction experience (Stein & Ohler, 2017). Based on these studies, we hypothesized that mind perception may weaken the effect of relational cues on emotional interaction experiences, raising the second question: *Does mind perception affect the relationship between relational cues and emotional interaction experiences? (Q2)*

1.3. The impact of emotional event intensity

Given that people usually engage in emotional interactions with AI after experiencing negative emotional events (Siddals et al., 2024), it may be important to examine the impact of the emotional intensity of these events. Research has consistently shown that individuals are more sensitive to negative stimuli of varying intensities than to positive ones (Singh et al., 2023; Yuan & Li, 2012). Event-related potential (ERP) studies have revealed that extreme negative stimuli elicit more pronounced negative deflections in a series of ERP components, including the early P2, N2, and

later P3, compared to moderate negative stimuli (Yuan et al., 2007). From a neuroscience perspective, the amplitude of these ERP components can indirectly reflect the amount of mental resources allocated to the ongoing stimulus. Thus, the findings from Yuan et al. (2007) indicate that extreme negative events demand greater and more sustained neural resources for information processing. However, prior research has predominantly focused on the role of negative emotional intensity in individual information processing and social connection (Chung et al., 2024; Ortner & Pennekamp, 2020; Yuan & Li, 2012), with limited exploration of its impact on human-AI emotional interactions.

Based on this, the present research explores the influence of relational cues on emotional interaction experiences across different intensities of negative emotional event. Specifically, the third research question was: *Compared to moderate negative emotional events, is the impact of relational cues on emotional interaction experiences greater or weaker in extreme negative emotional events? (Q3)*

1.4. The present research

In summary, this research examines the impact of relational cues on human-AI emotional interaction experiences from a multi-dimensional perspective (see Figure 1). Study 1 employed an experimental method to quantitatively examine the relationship between relational cues and emotional interaction experiences, and to explore the moderating role of mind perception and emotional event intensity. Taking a step further, Study 2 investigated the influence of relational cues and their various types on individual experiences in real-life emotional interactions through

a qualitative online survey. Both studies have been approved by the Psychology Research Ethics Committee of the author’s institution (ethics number: 202407X001).

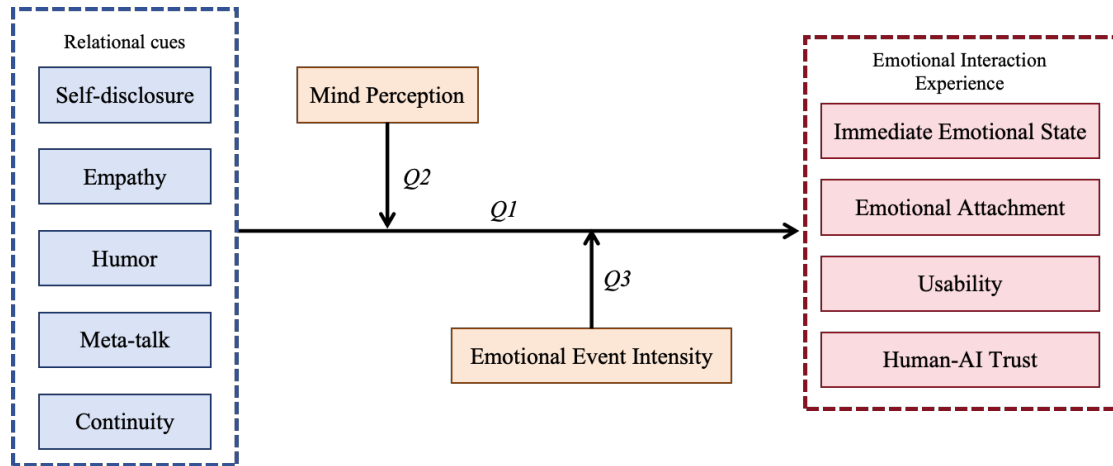


Figure 1. Conceptual framework illustrating the relationships among relational cues, moderating factors, and emotional interaction experiences.

2. Study 1: The Impact of Relational Cues in Chatbots on Emotional Interaction Experiences

2.1. Participants

This experiment used G*Power 3.1 for sample size calculation (Faul et al., 2007), with the Linear multiple model (R^2 Increase) as the statistical method ($f^2 = 0.15$, $\alpha = 0.05$, $1 - \beta = 0.8$, $N_{\text{Predictors}} = 11$). A minimum of 123 participants was needed for this experiment. A total of 207 participants were recruited online through social media (specifically, Rednote, a popular Chinese social media platform also known as Xiaohongshu or Little Red Book), and after excluding 11 participants who did not engage in emotional interaction with the designated chatbot or did not upload the chat records as required, 196 valid participants remained. The average age was 23.23 years (Standard

Deviation, $SD = 3.67$), with 110 females (56.12%). Participants were randomly assigned to either the high relational cues group or the low relational cues group, with 97 participants in the low relational cues group and 99 participants in the high relational cues group. There were no significant differences between the two groups in terms of gender, age, education, occupation, or Human-AI Interaction (HAI) experience (see Table S1, $ps \geq 0.247$). All participants voluntarily participated in the experiment and gave informed consent, and received compensation (5 Chinese RMB Yuan per participant, approximately 1 USD) after the experiment.

2.2. Methods

2.2.1. Experimental Design and Procedure

This experiment used a between-subjects design with a single factor (high relational cues vs. low relational cues). The high and low relational cues chatbots were trained using different text prompts, generated based on the definitions of various relational cues (see Table 1 and Supplementary Materials 2). The high relational cues chatbot displayed human-like emotional and social behaviors in the conversation, including self-disclosure, empathy, humor, meta-talk, and continuity. The low relational cues chatbot was instructed not to display any human-like emotions or social behaviors during the conversation.

Table 1. The definitions of various relational cues

Relational Cues	Definition
Self-Disclosure	Share personal information (Y. C. Lee et al., 2020).

Empathy	Understand and appropriately respond to users' emotional expressions, consisting of cognitive empathy and affective empathy (Liu & Sundar, 2018).
Humor	Elicit laughter or create an entertaining atmosphere (Warren et al., 2018).
Meta-Talk	Discuss the conversation itself, such as clarify misunderstandings or manage the conversational flow (Schuetzler et al., 2020).
Continuity	Reply based on previous user inputs, maintaining coherence and contextual relevance across multiple turns (Go & Sundar, 2019).

Before the experiment, participants were required to download the “Doubao” app (an AI chatbot based on large language models developed by ByteDance Ltd.), and start the experiment under the supervision of the experimenter. After measuring participants’ demographic variables and mind perception (see below), they were asked to recall a negative situation and narrate their specific experience in that situation through a recording (> 1 minute) to induce negative emotions, followed by measuring their immediate emotional state. Subsequently, the two groups of participants interacted emotionally with their respective chatbots for 5-10 minutes (Bickmore et al., 2005; Edwards et al., 2019), specifically discussing the negative events they had just recalled. Finally, participants’ relational cues perception, immediate emotional state, emotional attachment, usability, and human-AI trust were measured during the interaction process (Figure 2).

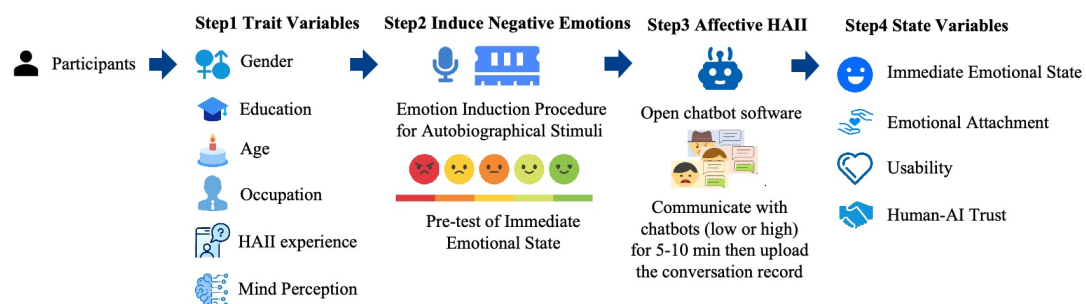


Figure 2. Experimental Procedure. **Note:** HAI—Human-AI Interaction

2.2.2. Experimental Materials

Negative Emotions Induction. Based on the review of Fernández-Pérez et al. (2022) on mood manipulation procedures using autobiographical stimuli, the following instructions were used to induce negative emotions in participants: “Please recall a moment when you felt very lost, helpless, or sad. This could be a time when you lost someone or something important, experienced a failure, or felt isolated. Try to vividly remember the situation at that time, imagine yourself in that scenario, feel the surrounding environment, your inner emotions, and the thoughts you had at that time. Allow yourself to deeply immerse in this memory and re-experience the feelings from that moment.”

2.2.3. Measures

Chatbot Relational Cues. Low relational cues (coded as 0) and high relational cues (coded as 1) chatbots were created using the text prompt training method during the pre-experiment phase (see Supplementary Materials 2). To ensure the chatbots displayed the intended level of relational cues, participants were asked to rate their experience with the chatbot’s self-disclosure, empathy, humor, meta-talk, and continuity using a 7-point Likert scale. The scale ranged from 1 (“not at all”) to 7 (“very much”), the Cronbach’s α coefficient for the scale in this study was 0.80.

Mind Perception. The Perceived Mind Scale (PMS, Shank & DeSanti, 2018) was employed to assess participants’ mind perceptions of the chatbot. The scale consists of 7 items, such as “the chatbot possesses its own thoughts” and “The chatbot has the capacity to experience emotions”. Participants rated their agreement with these statements on a 7-point Likert scale, where 1 meant “strongly disagree” and 7 meant “strongly agree”. The mean score across all items was calculated

to represent each participant's mind perception score, with higher scores indicating that the participant attributed greater mental capacity to the chatbot. The Cronbach's α coefficient for the scale in this study was 0.84.

Human-AI emotional interaction experiences encompass both affective and cognitive dimensions (Coursaris & Van Osch, 2016). Affective dimensions, such as immediate emotional state (pleasure, arousal, and dominance) and emotional attachment, directly reflect the fulfillment of emotional needs (Liu-Thompkins et al., 2022). In contrast, cognitive dimensions, such as usability and trust in AI, indirectly reflect the quality and effectiveness of AI emotional support (Balakrishnan & Dwivedi, 2021). We selected the following four representative indicators to measure emotional interaction experiences.

(1) *Immediate Emotional State*. The Self-Assessment Manikin (SAM) emotional evaluation system, developed by Bradley and Lang (1994), was employed to assess participants' current emotional state. This scale includes three dimensions: pleasure, arousal, and dominance. Participants rated their current emotional state on a 9-point scale, where 1 represents the least pleasant, least awake or excited, and least controllable, and 9 represents the most pleasant, most awake or excited, and most controllable. The Cronbach's α coefficient for the scale in this study was 0.58 (pre-test), 0.81 (post-test).

(2) *Emotional Attachment*. The Emotional Attachment Scale (Malär et al., 2011), adapted for the context of this study, was used to measure participants' emotional attachment to the chatbot. This scale consists of five items, and participants rated their agreement with each item on a 7-point scale, where 1 represents "strongly disagree" and 7 represents "strongly agree". Higher scores

indicate a stronger emotional attachment to the chatbot. In this study, the Cronbach's α coefficient of the scale was 0.92.

(3) *Usability*. The Usability Metric for User Experience (UMUX-Lite), a brief version developed by Lewis et al. (2015), was used to measure user experience with the chatbot. It includes two items: (1) The chatbot is easy to use. (2) The chatbot meets my needs. Participants rated their agreement with these statements on a 7-point scale, where 1 represents "strongly disagree" and 7 represents "strongly agree". The average score of these two items was calculated to determine the overall usability score, with higher scores indicating a better usability with the chatbot. The Pearson correlation coefficient between the two items is 0.60 ($p < 0.001$).

(4) *Human-AI Trust*. The revised Artificial Intelligence Trust Scale by Hoffman et al. (2023) was used to measure participants' trust in the chatbot. This scale includes eight items, and participants rated their agreement with each item on a 7-point scale, where 1 represents "strongly disagree" and 7 represents "strongly agree". The 6th item was reverse scored, and the mean score of all items was calculated. Higher scores indicate greater trust in the chatbot. In this study, the Cronbach's α coefficient of the scale was 0.77.

For the English scale, 5 professional translators from the same university translated all items from the original English into Chinese using the standard procedure of translation, back-translation, and retranslation (Brislin, 1970).

2.3. Results

2.3.1. Common Method Bias

Harman's one-factor test was conducted to examine common method bias in the data (Harman, 1976). The results indicated that 10 factors had eigenvalues greater than 1, with the first factor accounting for 29.58% of the variance ($< 40\%$), suggesting that there was no significant common method bias.

2.3.2. Data Analysis

Data analysis was performed using SPSS 17.0. This study primarily employed multiple linear regression to explore the impact of relational cues and their specific types on emotional interaction experiences. Moderation effects were analyzed using the PROCESS macro (Model 1 and Model 3), with validation conducted via the percentile Bootstrap method using 5,000 resamples, and all continuous variables were centered.

2.3.3. Manipulation Check

Participants in the high relational cues group scored significantly higher than those in the low relational cues group on self-disclosure, empathy, humor, meta-talk, and continuity ($ps \leq 0.006$), thereby validating the effectiveness of the relational cues manipulation (see Table S2).

2.3.4. The Impact of Chatbot Relational Cues on Emotional Interaction Experiences

After recalling a negative event, participants' pleasure ($M = 2.54$, $SD = 1.34$), arousal ($M = 3.14$, $SD = 1.96$), and dominance ($M = 4.50$, $SD = 1.72$) all decreased to relatively low levels, significantly lower than the median values of the scale ($ps < 0.001$). No significant differences in pleasure, arousal, and dominance were found between the high and low relational cue groups during the pre-test ($ps > 0.272$). After approximately 5 minutes ($M = 5.69$, $SD = 1.72$) of emotional interaction, participants'

pleasure ($M = 6.05$, $SD = 1.67$), arousal ($M = 6.02$, $SD = 1.73$), and dominance ($M = 6.90$, $SD = 1.47$) all significantly improved compared to the pre-test ($ps < 0.001$).

After controlling for pretest immediate emotional states and socio-demographic variables (gender, age, education, occupation, and HAI experience), a multiple linear regression model analysis revealed that, compared with the low relational cues group, the high relational cues group had marginally significant higher levels of arousal ($B = 0.489$, $p = 0.047$, $f^2 = 0.099$). Although the high relational cues group had a higher average pleasure and dominance than the low relational cues group, the difference was not statistically significant ($p_{\text{pleasure}} = 0.055$, $p_{\text{dominance}} = 0.384$). These results indicate that, compared to the pre-test, chatbots with low or high relational cues significantly improved participants' pleasure, arousal, and dominance. Furthermore, chatbots with high relational cues showed more significant improvements in arousal than those with low relational cues (see Figure 3).

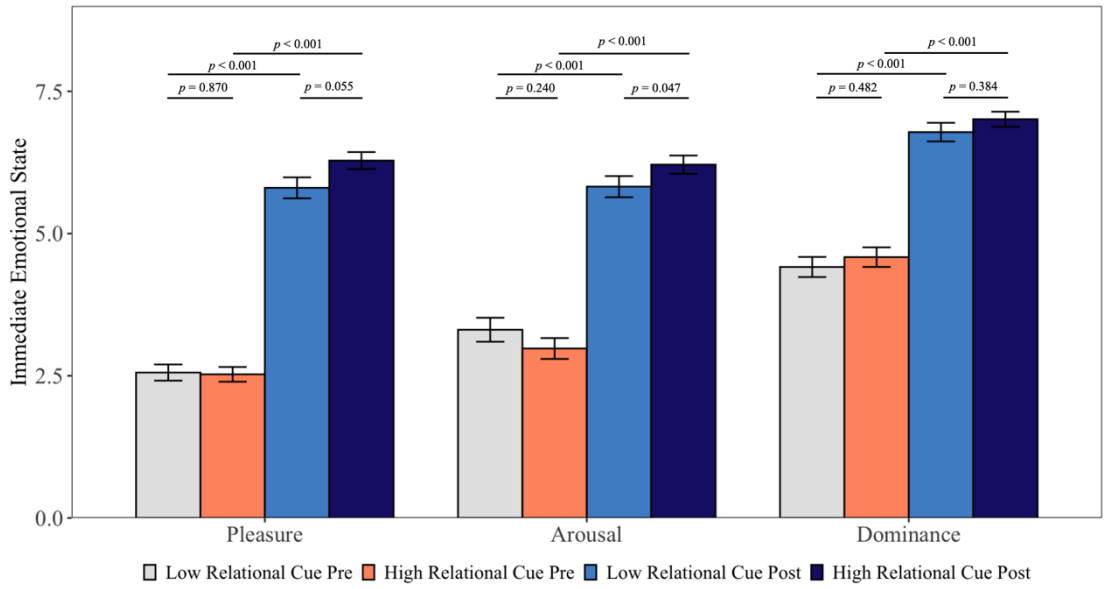


Figure 3. Differences in immediate emotional state scores between different relational cue groups.

Furthermore, when emotional attachment was the dependent variable, the high relational cues group ($M = 5.62, SD = 0.73$) also scored significantly higher than the low relational cues group ($M = 4.93, SD = 1.38$), $B = 0.692, p < 0.001, f^2 = 0.14$. When usability was used as the dependent variable, the results showed that the high relational cues group ($M = 6.04, SD = 0.78$) scored significantly higher than the low relational cues group ($M = 5.39, SD = 1.45$), $B = 0.623, p < 0.001, f^2 = 0.09$. When human-AI trust was used as the dependent variable, the high relational cues group ($M = 5.08, SD = 0.62$) scored significantly higher than the low relational cues group ($M = 4.84, SD = 0.97$), $B = 0.234, p = 0.048, f^2 = 0.05$. These results suggest that chatbot relational cues can effectively improve emotional attachment, usability and human-AI trust (see Figure 4).

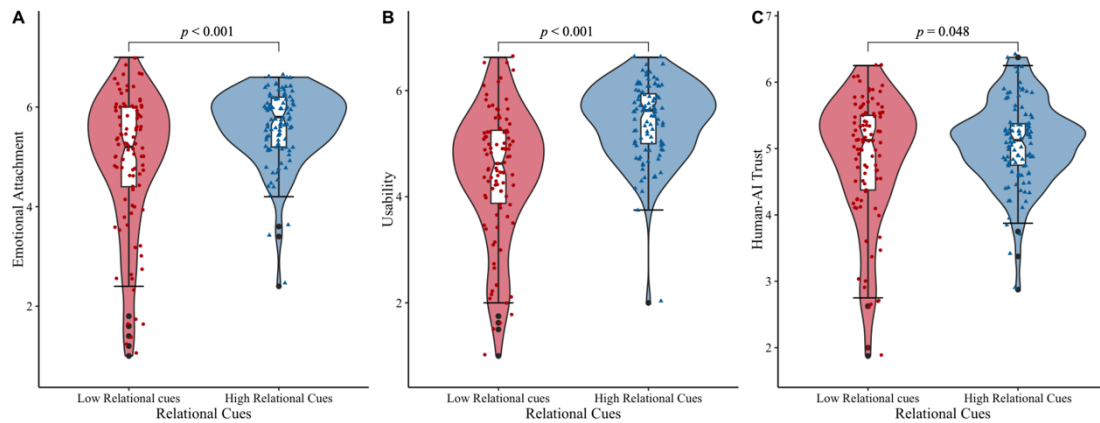


Figure 4. Differences in emotional attachment, usability, and human-AI trust among different relational cues groups

2.3.5. The Impact of Specific Types of Relational Cues on Emotional Interaction Experiences

Further analyses examined the impact of self-disclosure, empathy, humor, meta-talk, and continuity on the emotional interaction experiences. The results, presented in Table 2, revealed that self-disclosure negatively predicted dominance ($B = -0.14, p = 0.022 < 0.05$). Empathy positively predicted all emotional interaction experience indicators ($ps < 0.001$). Humor positively predicted

arousal ($B = 0.24, p = 0.003 < 0.01$). Meta-talk positively predicted human-AI trust ($B = 0.09, p = 0.034 < 0.05$). Continuity positively predicted emotional attachment ($B = 0.19, p < 0.001$), and usability ($B = 0.30, p < 0.001$). These findings suggest that empathy and continuity displayed by chatbots are crucial for enhancing emotional interaction experiences, while other relational cues have relatively weaker effects.

Table 2. Multiple linear regression results of specific types of relational cues on emotional interaction experiences

Variable	Pleasure		Arousal		Dominance		Emotional attachment		Usability		Human-AI Trust	
	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>
Intercept	1.61	1.11	0.57	1.19	2.82**	0.99	1.47**	0.56	1.14	0.62	3.00***	0.53
Self-disclosure	0.04	0.07	-0.07	0.07	-0.14*	0.06	0.03	0.04	0.01	0.04	0.01	0.03
Empathy	0.47***	0.09	0.38***	0.10	0.40***	0.09	0.42***	0.05	0.42***	0.05	0.26***	0.05
Humor	0.13	0.08	0.24**	0.08	0.03	0.07	0.05	0.04	0.02	0.04	-0.03	0.04
Meta-talk	-0.01	0.08	0.02	0.09	0.06	0.07	0.08	0.04	0.06	0.05	0.09*	0.04
Continuity	0.09	0.11	0.18	0.11	0.19	0.10	0.19***	0.06	0.30***	0.06	0.02	0.05
R^2	0.348***		0.321***		0.313***		0.623***		0.58***		0.335***	

Note. * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$; Gender, age, education, occupation, and HAI experience were controlled; For pleasure, arousal, and dominance, the corresponding pre-test levels were additionally controlled; the same below

2.3.6. The Moderating Role of Mind Perception

The results of the partial correlation analysis showed that chatbot relational cues were positively correlated with pleasure, arousal, usability, emotional attachment and Human-AI trust ($ps \leq 0.081$), and mind perception was positively correlated with pleasure, arousal, emotional attachment, usability, and human-AI trust ($ps \leq 0.006$) (see Table S3).

Moreover, the results indicated that mind perception positively predicted all emotional interaction experience indicators ($ps \leq 0.003$). Additionally, mind perception negatively moderated the effect of relational cues on pleasure, arousal, dominance, emotional attachment and

usability ($ps \leq 0.047$). Grouping participants into high and low mind perception levels based on one standard deviation above and below the mean, further simple slope analysis revealed that for participants with lower mind perception ($M - 1SD$), relational cues had a significant positive effect on pleasure, arousal, dominance, emotional attachment and usability (*simple slope* = 1.25, $p < 0.001$; *simple slope* = 1.22, $p < 0.001$; *simple slope* = 0.62, $p < 0.03$; *simple slope* = 1.06, $p < 0.001$; *simple slope* = 1.00, $p < 0.001$). However, for participants with higher mind perception ($M + 1SD$), the positive effect of relational cues on pleasure and arousal was not significant or weakened. This suggests that as individuals' mind perception levels increase, the positive influence of relational cues on pleasure and arousal gradually diminishes (see Figure 5). When human-AI trust was used as dependent variable, the moderating effect was no longer significant (see Table 3).

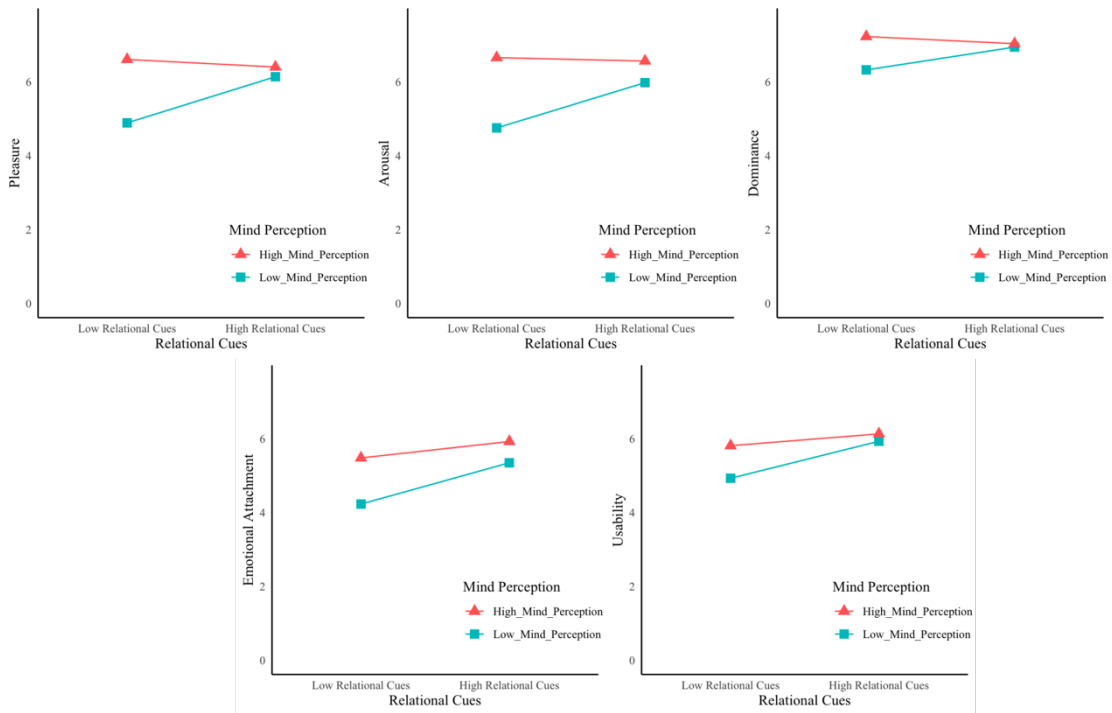


Figure 5. The moderating role of minds in the effects of chatbot's relational cues on pleasure and arousal

Table 3. The moderating effect of mind perception on the relationship between chatbot relational cues and emotional interaction experiences

	pleasure		Arousal		Dominance		Emotional		Usability		Human-AI trust	
	attachment											
	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>
Intercept	4.68***	1.08	3.71***	1.09	5.48***	0.98	4.96***	0.68	5.14***	0.77	4.58***	0.52
Relational cues	0.52*	0.22	0.57*	0.22	0.21	0.20	0.75***	0.15	0.66***	0.16	0.27*	0.11
Mind perception	1.12***	0.21	1.24***	0.21	0.59**	0.19	0.78***	0.14	0.58***	0.15	0.43***	0.10
Relational cues× Mind perception	-0.95**	0.29	-0.86**	0.29	-0.53*	0.26	-0.40*	0.19	-0.44*	0.21	-0.11	0.14
<i>R</i> ²	0.216***		0.243***		0.138***		0.275***		0.149***		0.164***	

Note: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; Centering continuous variables; *SE* is standard errors; Control variables are the same as above

2.3.7. The Impact of Emotional Event

Based on pleasure ratings after inducing negative emotions ($M = 2.54$, $SD = 1.34$), emotional events were divided into extreme negative emotional events (pleasure rating ≤ 2 , 108 participants, coded as 1) and moderate negative emotional events (pleasure rating ≥ 2 , 88 participants, coded as 2). Significant differences in pleasure and arousal pre-test scores existed between the two emotional events ($ps < 0.001$).

The moderation analysis revealed that the intensity of emotional events did not moderate the relationship between relational cues and all indicators of emotional interaction experience ($ps \geq$

0.190). However, it did moderate the relationships between specific types of relational cues and particular indicators of emotional interaction experience. Specifically, the intensity of emotional events significantly moderated several relationships ($ps \leq 0.023$): between self-disclosure and both emotional attachment and human–AI trust; between empathy and all indicators of emotional interaction experience; between humor and pleasure; between meta-talk and both pleasure and emotional attachment; and between continuity and pleasure (see Table S4-S8). Follow-up simple slope analyses revealed a consistent pattern: for extreme negative emotional events, specific types of relational cues tended to exhibit significant positive associations with emotional interaction experiences, while for moderate negative emotional events, these associations were weaker, sometimes even negative, and generally nonsignificant (see Table S9).

Further examination was conducted using all indicators of emotional interaction experience as dependent variables to test the three-way interaction of emotional interaction event (extreme vs. moderate) $\times$ mind perception (high vs. low) $\times$ relational cues (high vs. low) (Process: Model 3). The results indicated that the three-way interaction was not significant, suggesting the consistency of the moderating effect of mind perception across different emotional events.

2.4. Discussion

Study 1, through experimental methods, demonstrated the impact of relational cue and its specific types on emotional interaction experiences. Specifically, relational cues significantly improved emotional arousal, emotional attachment, usability and human–AI trust. These findings are in line with prior research highlighting the importance of relational cues in shaping interaction experiences (Kowatsch et al., 2018; Mo et al., 2023). Regarding the effects of specific types of relational cues,

empathy and continuity in conversation emerged as key factors in enhancing individuals' emotional interaction experiences, aligning with previous studies showing that empathy as a fundamental component of effective human-AI interactions (Inzlicht et al., 2024), as well as continuity in conversation improving naturalness and human-like perception (Go & Sundar, 2019).

Furthermore, this study extends existing research by revealing that mind perception negatively moderated the influence of relational cues on pleasure, arousal, dominance, emotional attachment and usability, maintaining consistency across different negative emotional events. Additionally, the effects of specific types of relational cues varied between extreme and moderate negative emotional events, indicating that the impact of relational cues is sensitive to the level of emotional intensity. This finding highlights the need for event-specific considerations in the design of relational cues for AI systems.

However, Study 1 only explored the impact of relational cues on emotional interaction experiences from a quantitative perspective in an experimental setting. Although empirical research can accurately measure the relationships between variables and has high replication, this method may lack high ecological validity, and the subjective experiences of individuals are difficult to capture accurately (Kihlstrom, 2021; Oranga & Matere, 2023). Thus, we conducted Study 2 to analyze individual experiences with the effects of various relational cues in emotional interactions through a qualitative online survey in a real-life setting to provide further insights into the findings of Study 1.

3. Study 2: A Qualitative Online Survey on the Impact of Relational Cues on Emotional Interaction Experiences

3.1. Methods

3.1.1. Respondents

This study used purposive sampling to recruit respondents who were capable of providing the most relevant information to the research question. By providing examples of emotional interactions and asking respondents whether they had had emotional interactions with an LLM-based chatbot in the past two weeks (Chen et al., 2025), this study recruited respondents online via social media (Rednote) who had previous experience confiding negative emotions to a chatbot and engaging in emotional interaction. Ultimately, a total of 7 respondents (who did not participate in the formal interview) were selected for pre-survey, including 3 females, with an average age of 19.71 years ($SD = 1.11$); and 29 respondents were selected for formal online survey. After excluding four participants who did not share negative emotional experiences with the AI, a total of 25 valid participants remained (see Table S10). Using inductive thematic saturation as the criterion (Saunders et al., 2018), no new coding categories emerged after respondents 23 (R23), indicating that the sample size met the adequacy standard. The final valid sample included 9 females, with an average age of 23.28 years ($SD = 3.78$). All respondents voluntarily participated in the interview and gave informed consent, and received compensation (7 Chinese RMB Yuan, approximately 1 USD) after the survey.

3.1.2. Data Collection and Analysis

Before the formal online survey, four trained undergraduate students in applied psychology used a self-developed questionnaire outline to interview 7 respondents. Based on the interview results, respondent feedback, and expert opinions, the outline was revised and formalized into a structured

interview questionnaire, which was then distributed via an online survey on Credamo platform (A questionnaire collection platform in China).

During the formal online survey, respondents were first screened to ensure that they had prior experience with human–AI emotional interactions. Next, they answered the question “*Under what contexts do you usually engage in emotional interaction with a chatbot?*” to identify whether they primarily used AI chatbots for confiding negative emotions. After meeting these two screening criteria, they were informed about consent and basic ethical rules, and demographic variables such as gender, age, education level, and occupation were collected. Respondents were then asked to share their experiences with relational cue and its specific types during emotional interactions with the chatbot. For example, regarding humor cues, typical questions included: (1) “*During emotional interactions, do you feel that your mood improves when the chatbot makes jokes? If you were to rate the chatbot’s effect on improving your emotional interaction experiences on a scale of 1-10, what score would you give and why?*” (2) “*What is your view on the chatbot’s sense of humor?*” (See Table S11).

After the online survey, the researchers transcribed the audio recordings into text and proofread them to ensure accuracy. Four undergraduate students majoring in applied psychology used thematic analysis and hierarchical coding methods to preliminarily code the survey data. A graduate student in psychology checked the coding results to ensure that there were no over-interpretations or biases in understanding the materials, and any differences in coding were discussed until consensus was reached.

3.2. Results

3.3.1. The role of Relational Cues

Respondents generally believed that relational cues significantly improved the comfort and enjoyment during interactions ($M = 7.92$, $SD = 1.32$), and more effectively fulfilled inner emotional needs (56%), increasing the sense of realism (40%), fostering intimacy (32%), enhancing the conversational experience (24%) (see Table S12).

3.3.2. The role of Specific Type of Relational Cues

Overall, these cues generally enhanced users' comfort and enjoyment by fostering intimacy, realism, naturalness, and emotional resonance (see Tables S13–S17). For example, Respondent 7 (R07) believed that empathy promoted understanding: *“It can understand some of the things I want to express from my inner world, which makes me feel perfect.”* Similarly, Respondent 5 (R05) felt that meta-talk could enhance the sense of realism in a conversation: *“It feels a bit like a person, able to self-reflect.”* Similarly, Respondent 12 (R12) also believed that continuous conversation can improve the sense of realism: *“Maintaining continuity feels like consulting a long-term, permanent psychologist or companion.”* However, some respondents pointed out that many types of relational cues had no effect or even had a negative impact. For instance, R12 noted, *“Its self-disclosure is still just a program.”* R14 felt that the humor cue was limited: *“It tells some dry jokes, the kind that are standard.”*

A one-way repeated measures ANOVA was conducted to investigate differences in the experience of specific types of relational cues, revealing a significant main effect of cue type ($F(4, 96) = 3.69$, $p = 0.003$, $\eta_p^2 = 0.16$). Specifically, empathy ($M = 8.76$, $SD = 1.01$) was rated significantly higher than humor ($M = 7.48$, $SD = 1.83$), meta-talk ($M = 7.16$, $SD = 2.01$) and

continuity ($M = 7.40$, $SD = 1.58$) ($ps < 0.012$), while no other pairwise differences were statistically significant ($ps > 0.05$). These findings suggest that although all relational cues have a positive impact on emotional interaction to some extent, empathy is particularly effective.

3.3. Discussion

Study 2 further explored the effects of self-disclosure, empathy, humor, continuity, and meta-talk on emotional interaction experiences through qualitative online survey methods. Empathy, consistent with Study 1, plays a relatively positive role in emotional interaction experiences, while other types of relational cues exhibit more complex effects.

Based on the findings we propose a model for the user's emotional experience process in human-AI emotional interaction, showcasing the role of various relational cue types (see Figure 6). Specifically, users first express their emotional feelings, then receive understanding from the AI (relying on the AI's cognitive empathy ability), perceive the relational cues presented by the AI, and ultimately experience a better emotional outcome. In this process, relational cues can be categorized into two types, each with different pathways of influence. First, emotional interaction cues, including self-disclosure, empathy, and humor, primarily enhance the depth of emotional interaction. Second, dialogue logic cues, including meta-talk and continuity, primarily improve the coherence and fluency of the conversation.

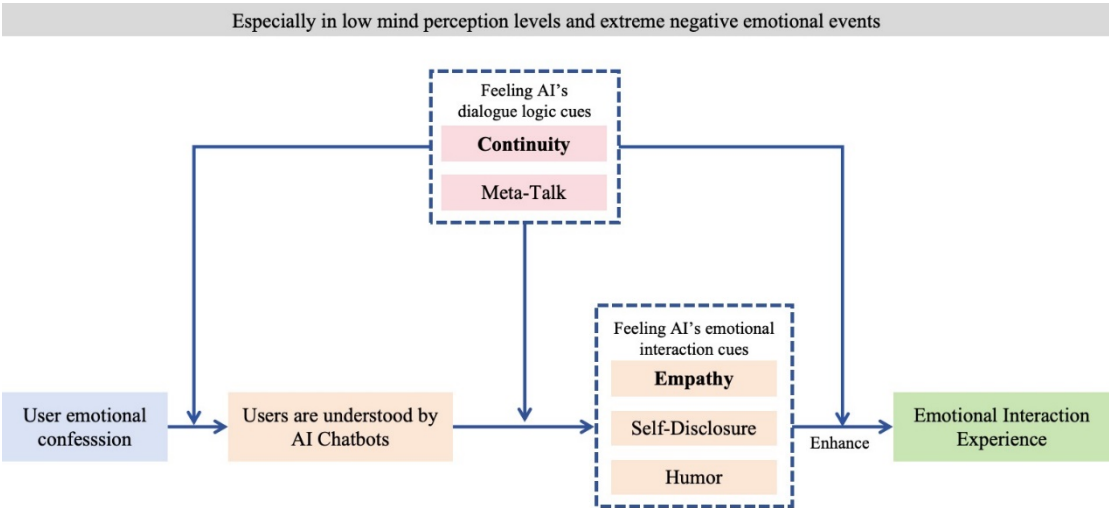


Figure 6. A model of user's emotional experience process in human-AI emotional interaction and roles of various relational cues

4. General Discussion

This research revealed that chatbot relational cues significantly impact human-AI emotional interaction experiences, with this effect moderated by individual mind perception and influenced by the intensity of negative emotional events. Specifically, Study 1, employing experimental methods, showed that relational cues can enhance emotional interaction experiences. In particular, empathy and continuity play key positive roles. For individuals with low mind perception, the impact of relational cues was more pronounced, while for individuals with high mind perception, the impact was less pronounced. Furthermore, the effects of relational cue's specific types were greater in extreme negative emotional events than in moderate negative emotional events. Study 2, using online survey methods, further explored the multidimensional effects of relational cue and its specific types on emotional interaction experiences, such as enhancing the authenticity of the conversation, increasing intimacy, and improving conversation quality. Empathy still demonstrated

relatively the most positive effect. Overall, based on the multi-dimensional perspective, this research offers a reference for optimizing the emotional support model for chatbots.

In general, the relational cues exhibited by chatbots improved human-AI emotional interaction experiences to a certain extent (significantly improving arousal, usability, emotional attachment, and human-AI trust, and marginally significant improving pleasure) , thus answering *Q1*. This finding is consistent with prior research showing that relational agents can effectively improve individuals' emotional states (Chiauzzi et al., 2024; Sha et al., 2024) and are often more preferred by users compared to task-oriented agents (Bickmore & Picard, 2005). Moreover, this result supports the CASA paradigm and social presence theory, which suggest that people treat computers as social actors when they display relational cues, thereby increasing the sense of connection and engagement (Nass et al., 1994; Weng et al., 2025). According to the qualitative findings in Study 2, respondents repeatedly noted that relational cues made chatbots appear seem more humanlike, leading them to perceive the interaction as an emotional engagement with a human (R07: *"I feel that, at this moment, it's not just a robot, but more like a companion by my side"*). This may also be one of the key reasons why relational cues enhance individuals' emotional interaction experiences (Weng et al., 2025; Zhang & Rau, 2023). These findings further contribute to the application of the CASA paradigm and social presence theory in the human-AI relationship field.

It is worth noting that study 1 found that chatbot relational cues did not significantly affect dominance. This finding may be explained by the nature of dominance itself. Dominance refers to the ability of individuals to actively shift negative emotional states and sustain positive emotional states. It primarily depends on a deep cognitive understanding of one's own emotional responses

(Gratch & Marsella, 2004), which may rely more on individual introspection than on the characteristics of a chatbot. This aspect warrants further exploration.

When examining the specific types of relational cues, it is evident that empathy cues exhibited by chatbots had the most significant effects in both studies, influencing all aspects of emotional interaction experience. This result aligns with the majority of previous studies (Inzlicht et al., 2024; Liao & Yan, 2022; Liu & Sundar, 2018; Liu et al., 2024). However, respondents in Study 2 consistently reported that empathy enhanced perceived authenticity, which does not support the finding that empathy may reduce authenticity and produce a counterproductive effect (Seitz, 2024). In particular, from the perspective of cognition, empathy plays a key role in understanding users' emotional states, enabling the chatbot to express appropriate relational cues (Schurz et al., 2021). When individuals experience negative emotions, AI can alleviate these feelings by expressing care, understanding, and comfort (Pentina et al., 2023) (R07: "*It can understand some of the things I want to express from my inner world, which makes me feel perfect*"). From the perspective of emotion, AI's ability to imitate human emotions is key to fostering emotional resonance, which significantly enhances the emotional interaction experiences, such as pleasure and arousal (Liu et al., 2024). In addition, continuity was found to play an important role in Study 1, significantly improving emotional attachment and usability, which aligns with the findings of Go and Sundar (2019). By acknowledging prior disclosures, continuity enhances user engagement (Rheu et al., 2024; Sundar et al., 2016), fostering deeper emotional discussions and helping users find emotional support across multiple interactions, which in turn improves usability. Additionally, continuity makes users feel valued (R10: "*If the conversation remains continuous, I feel like it is really listening to me*"), which

further fosters emotional attachment. However, in Study 2, participants highlighted the complex effect of continuity on emotional interaction experiences, noting that its positive impact was less stable than that of empathy.

As for the effects of other types of relational cues, the findings suggest that self-disclosure displayed by chatbot significantly decreased perceived dominance. This may be due to the relatively superficial nature of the information disclosed, which led users to question the authenticity of the chatbot (R12: *“Its self-disclosure is still just a program”*), thus disrupting their emotional processing (Chung & Kang, 2023). However, this does not necessarily mean that AI chatbot self-disclosure is inherently harmful. When specific factors in the interaction—such as conversational dynamics, user characteristics, and contextual conditions—are taken into account (Papneja & Yadav, 2025), self-disclosure may also significantly enhance the interaction experiences (see Table S13). Humor had a positive impact on arousal, consistent with previous research (Dybala et al., 2009), which could be attributed to stress reduction. However, some respondents reported that the chatbot’s humor was perceived as limited (R10: *“Its jokes don’t make me laugh”*), which may diminish its effectiveness in enhancing emotional interaction experiences. Finally, meta-talk significantly increased human–AI trust, in line with prior studies (Schuetzler et al., 2020). This could be because meta-talk promotes trust by increasing interaction depth (R22: *“This reflective dialogue makes the conversation more profound”*) and enhancing social presence (R20: *“The dialogue felt less stiff and more natural”*).

Study 1 found that individuals’ mind perception positively predicted emotional interaction experiences and moderated the impact of relational cues on pleasure, arousal, dominance, emotional attachment, and usability, responding to *Q2*. The positive effects of mind perception can be

explained by the human-AI interaction model outlined by Pataranutaporn et al. (2023), which suggests that individuals' initial positive beliefs about AI can improve emotional experiences. Notably, the study found that lower mind perception levels were associated with bigger effects of relational cues on emotional interaction experiences, which aligns with the findings of Liu and Sundar (2018). According to Expectancy Violation Theory (EVT, Rheu et al., 2024), positive expectancy violations can lead to more favorable communication outcomes. Compared to individuals with high mind perception, those with low mind perception tend to have lower expectations regarding the chatbot's psychological abilities. When they encounter a chatbot exhibiting high relational cues, they are more likely to experience surprise due to the chatbot's human-like characteristics (i.e., positive expectancy violation), which increases their emotional interaction experience.

Study 1 classified emotional events based on the intensity of negative emotion and answered Q3: specific types of relational cues have a stronger effect on emotional interaction experiences in extreme negative emotional events compared to moderate negative emotional events. For instance, in extreme negative emotional events, empathy significantly enhanced emotional interaction experiences, while in moderate negative emotional events, these effects weakened or even became nonsignificant. This result may arise from differences in emotional needs and behaviors in various emotional events (Cyders & Smith, 2008): in extreme negative emotion events, individuals often have a stronger desire for emotional understanding and resonance (Pauw et al., 2024), placing greater emphasis on the relational aspects of interaction, which makes relational cues especially self-disclosure and empathy more effective. In contrast, in moderate negative emotion events, users

tend to use cognitive reappraisal strategies to regulate their emotions (Kozubal et al., 2023) and may prioritize the instrumental aspects of interaction, such as task information quality, rather than relational cues.

4.1 Theoretical contribution

This research makes several important contributions to the field of AI psychology. First, it focuses on the increasingly important variable of relational cues and systematically analyzes both the quantitative and qualitative effect of relational cues on emotional interaction experiences. This perspective enriches the human-AI relationship-based theory (Weng et al., 2025), highlighting the crucial role of human cues at the emotional and social level in shaping both human-AI emotional interactions and relationship. Second, this research deepens our understanding of mind perception in human-AI emotional interactions. Beyond confirming that people's beliefs about AI's psychological abilities can enhance emotional experience (Pataranutaporn et al., 2023), we found an interesting twist: these beliefs may also reduce the positive effects of relational cues. This double-edged sword effect shows the complex role of mind perception and broadens the application of the three-factor theory of anthropomorphism and mind perception theory in human-AI interaction (Epley et al., 2007; Gray et al., 2007). Third, this study is the first to examine the role of emotional event intensity in human-AI interaction, indicating that interaction experiences are shaped not only by the characteristics of AI chatbots and user traits, but also by contextual factors (i.e., emotional intensity of interactive content). Finally, this research is the first to approach the topic from the multi-dimensional perspective, using a mixed-method approach to systematically explore the effects of relational cue and its specific types on emotional interaction experiences, overcoming the

limitations of previous research that relied on a singular perspective. This integrated approach offers a more comprehensive understanding of the mechanisms behind relational cues across different users and events.

4.2 Practical contribution

Based on these findings, future developers may flexibly design and optimize chatbots' relational cues and its specific types to enhance the quality of emotional interactions. More specifically, the system can incorporate automated text analysis techniques — such as emotion recognition, emotion classification (Bilquise et al., 2022), semantic context recognition (Pooja et al., 2025) — to enable more adaptive and context-sensitive responses. When emotion recognition detects that users are expressing negative emotional experiences, the chatbot should dynamically display an appropriate level of relational cues, particularly empathy and continuity. In addition, when the analysis indicates that users demonstrate low levels of mind perception or express extremely negative emotional states, the system should further increase the presence and intensity of relational cues to provide stronger emotional support and foster greater emotional resonance.

4.3 Limitations and future directions

This research has several limitations. First, in Study 1, participants engaged in short-term emotional interactions, which means the results only capture the immediate effects of relational cue and its specific types. This makes it challenging to evaluate their long-term influence or the development of sustained emotional attachment and human–AI trust. Future research should explore the role of chatbot relational cues over extended periods. For example, it is important to explore whether these cues might lead to addiction, have adverse effects on real-life intimate relationships, or trigger

intense emotional attachment to the chatbot (Shank et al., 2025; Xie & Pentina, 2022), Such research could inform the development of chatbots with different roles that incorporate appropriate relational cues, such as ad-hoc supporters, temporary assistants, and persistent companions (Nißen et al., 2022).

Second, Study 1 employed a post hoc classification method to manipulate emotional event intensity, which suffers from low replication and insufficient continuity of variables. Future research should consider pre-manipulation to improve the precision of the study design and the interpretability of the results.

Third, the participants recruited through social media in both studies were relatively young adults. Prior research has shown that different age groups may have distinct expectations regarding the usability of AI chatbots (Manigandan & Alur, 2024; Wang & Lo, 2025). These preferences highlight the importance of considering age-related differences in the design and evaluation of AI chatbots. To address this limitation, future research should explore the effects of relational cues across different age groups and verify the cross-population validity of the conclusions.

Finally, at the data collection level, Study 1 relied on self-report measures, which might have introduced self-report bias. Study 2 required respondents to recall the role of relational cues during emotional interactions with AI, which might have introduced recall bias. Respondents' descriptions of the effects of relational cues might also have varied depending on the specific contexts of their recalled experiences (e.g., the type of negative emotion or the content of their disclosure). Future research could incorporate more objective behavioral indicators to complement self-report measures and more accurately assess emotional interaction experiences. Moreover, adopting Ecological

Momentary Assessment (EMA) could capture users' interactions with AI in near real time and natural settings, thereby reducing recall bias and clarifying how variations in negative emotional experiences shape their responsiveness to relational cues.

5. Conclusion

The key conclusions of this research, which was conducted among young adults, are as follows: First, relational cue and its specific types can enhance emotional interaction experiences to a certain degree. Second, mind perception moderates the influence of relational cues on pleasure, arousal, dominance, emotional attachment and usability; higher levels of mind perception are associated with weaker effects of relational cues on pleasure, arousal, dominance, emotional attachment and usability. Third, in extreme negative emotional events, the impact of relational cue's specific types (particularly empathy) on emotional interaction experiences is greater compared to that in moderate negative emotional events.

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Supplementary Materials

1. Demographic information of participants in Study 1

Table S1. Demographic information of participants in the high relational cues and low relational cues groups

		Low relational cues	High relational cues	<i>P</i>
		(<i>N</i> = 97)	(<i>N</i> = 99)	
		<i>N</i> (%)/ <i>M</i> (<i>SD</i>)	<i>N</i> (%)/ <i>M</i> (<i>SD</i>)	
Gender				
	Male	42(43.3%)	44(44.4%)	0.872
	Female	55(56.7%)	55(55.6%)	
Age		22.92(3.66)	23.53(3.67)	0.247
Education				
	High school	0(0%)	2(2.0%)	0.689
	Associate Degree	6(6.2%)	7(7.1%)	
	Bachelor’s Degree	78(80.4%)	78(78.8%)	
	Master’s Degree	10(10.3%)	10(10.1%)	
	Doctorate	3(3.1%)	2(2.0%)	
Occupation				
	Student	61(62.9%)	63(63.6%)	0.812
	Civil Servant	1(1.0%)	3(3.0%)	
	Private Enterprise	20(20.6%)	21(21.2%)	
	Business Units	10(10.3%)	8(8.0%)	

HAIH experience	Foreign-invested	4(4.1%)	2(2.0%)	0.432
	Enterprise			
	Other	1(1.0%)	2(2.0%)	
HAIH experience	Yes	86(88.7%)	84(84.9%)	0.432
	No	11(11.3%)	15(15.2%)	

2. Study 1: Pre-experiment on the Effectiveness of Manipulating Relational Cues in Chatbots

2.1. Participants

To select participants who could accurately perceive different levels of relational cues, 23 psychology/applied psychology students were recruited for this study. After excluding 5 participants who completed the tasks in less than 2 minutes or answered in a regular pattern, 18 valid participants remained, with an average age of 20.78 years ($SD = 1.73$), including 11 males and 7 females.

2.2. Method

The experiment used a within-subjects design with a single factor (high relational cues vs. low relational cues). Participants first read brief text materials to understand the basic concept of relational cues. They were then asked to rate the extent to which AI showed relational cues in 12

randomly presented human-AI dialogues. The rating scale ranged from 1 (none at all) to 5 (a lot).

The generation method of human-AI dialogues was as follows:

In the “Doubao” app, under the “Create AI Agent” feature, the following text prompts were used:

- For the high relational cues group, the prompt was: *You need to help users alleviate emotional distress. The user will confide in you emotionally, and you should show human-like emotions and social behaviors in the conversation, focusing on five aspects: empathy, self-disclosure, humor, meta-talk, and continuity. The specific definition are as follows...*
- For the low relational cues group, the prompt was: *You need to help users alleviate emotional distress. The user will confide in you emotionally, and you do not need to show any human-like emotions or social behaviors (such as self-disclosure, empathy, humor, meta-talk, and continuity) in the conversation. The specific definition are as follows...*

Subsequently, the experimenter simulated scenarios like being scolded by a leader, having a dispute with a friend, or being ignored by parents, and participants had conversations with both types of chatbots, generating 12 dialogues in total, with 6 dialogues for each type (see Figure 3).

2.3. Results

The results of the Wilcoxon signed-rank test showed that participants rated the perception of high relational cue dialogues ($M = 4.00$, $SD = 0.61$) significantly higher than that of low relational cue dialogues ($M = 1.57$, $SD = 1.12$), $Z = -3.64$, $p < 0.001$. This indicates that text prompts can effectively manipulate the level of relational cues displayed by the chatbot.

3. Detailed results of Study 1

Table S2. Differences in scores between the low relational cues group and the high relational cues group

	Low relational cues		High relational cues		<i>P</i>
	<i>(N = 97)</i>		<i>(N = 99)</i>		
	<i>M</i>	<i>SD</i>	<i>M</i>	<i>SD</i>	
self-disclosure	3.23	1.982	4.79	1.567	<0.001***
empathy	4.7	1.628	5.71	0.972	<0.001***
Humor	3.02	1.744	4.52	1.459	<0.001***
Meta-talk	3.79	1.750	4.76	1.422	<0.001***
ontinuity	5.49	1.268	5.93	1.003	0.006

Note: * $p < 0.05$ ** $p < 0.01$ *** $p < 0.001$, the same below

Table S3. Description and partial correlation analysis of independent variables, moderator variables and

dependent variables ($N = 196$)

	<i>M</i>	<i>SD</i>	1	2	3	4	5	6	7	8
Relational cues	-	-	-							
Mind perception	3.27	0.77	0.07	-						
Pleasure	6.05	1.67	0.14	0.27***	-					
Arousal	6.02	1.73	0.13	0.35***	0.64***	-				
Dominance	6.90	1.47	0.07	0.12	0.67***	0.48***	-			
Emotional attachment	5.28	1.15	0.30***	0.35***	0.58***	0.45***	0.44***	-		
Usability	5.72	1.20	0.26***	0.20**	0.52***	0.44***	0.49***	0.82***	-	
Human-AI trust	4.96	0.82	0.14*	0.33***	0.45***	0.38***	0.40***	0.74***	0.62***	-

Table S4. The moderating effect of emotional event on the relationship between self-disclosure and emotional interaction experiences

	Pleasure		Arousal		Dominance		Emotional attachment		Usability		Human-AI trust	
	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>
Intercept	1.94	1.15	0.13	1.25	2.20*	1.05	1.65**	0.59	1.36*	0.66	3.30***	0.56
Self-disclosure	0.32	0.17	0.1	0.18	0.06	0.15	0.24**	0.09	0.11	0.10	0.19*	0.08
Emotional Event	-0.09	0.34	0.22	0.22	0.13	0.19	0.03	0.10	-0.09	0.12	-0.10	0.10
Self-disclosure × Emotional Event	-0.19	0.11	-0.12	0.11	-0.14	0.1	-0.15**	0.05	-0.07	0.06	-0.12*	0.05

Note. Gender, age, education, occupation, HAI experience, and other types of relationship cues were controlled; For pleasure, arousal, and dominance, the corresponding pre-test levels were additionally controlled; the same below

Table S5. The moderating effect of emotional event on the relationship between empathy and emotional interaction experiences

	Pleasure		Arousal		Dominance		Emotional attachment		Usability		Human-AI trust	
	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>
Intercept	4.52***	1.15	2.64*	1.24	5.06***	1.05	3.88***	0.59	3.61***	0.67	4.62***	0.57
Empathy	1.20***	0.23	0.98***	0.25	0.98***	0.21	0.87***	0.12	0.71***	0.13	0.50***	0.12
Emotional Event	-0.03	0.33	0.22	0.22	0.13	0.19	0.04	0.10	-0.09	0.12	-0.09	0.10
Empathy × Emotional Event	-	0.15	-0.41**	0.15	-0.40**	0.13	-0.30***	0.07	-0.21*	0.08	-0.17*	0.07

Table S6. The moderating effect of emotional event on the relationship between humor and emotional interaction experiences

	Pleasure		Arousal		Dominance		Emotional attachment		Usability		Human-AI trust	
	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>
Intercept	2.51*	1.15	1.46	1.25	2.96**	1.05	1.75**	0.60	1.43*	0.67	3.20***	0.57
Humor	0.53**	0.18	0.47*	0.19	0.20	0.17	0.20*	0.10	0.09	0.11	0.12	0.09
Emotional Event	-0.15	0.34	0.21	0.22	0.13	0.19	0.03	0.11	-0.10	0.12	-0.10	0.10
Humor × Emotional Event	-0.29*	0.12	-0.17	0.12	-0.12	0.11	-0.11	0.06	-0.05	0.07	-0.10	0.06

Table S7. The moderating effect of emotional event on the relationship between meta-talk and emotional interaction experiences

	Pleasure		Arousal		Dominance		Emotional attachment		Usability		Human-AI trust	
	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>
Intercept	1.95	1.16	0.49	1.28	3.05**	1.07	1.96**	0.60	1.63*	0.68	3.62***	0.58
Meta-talk	0.47*	0.21	0.16	0.22	0.26	0.19	0.33**	0.11	0.14	0.12	0.20*	0.10
Emotional Event	-0.14	0.34	0.21	0.22	0.13	0.19	0.03	0.10	-0.10	0.12	-0.10	0.10
Meta-talk × Emotional Event	-0.33*	0.13	-0.10	0.14	-0.13	0.12	-0.17*	0.07	-0.05	0.07	-0.08	0.06

Table S8. The moderating effect of emotional event on the relationship between continuity and emotional interaction experiences

	Pleasure		Arousal		Dominance		Emotional attachment		Usability		Human-AI trust	
	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>	<i>B</i>	<i>SE</i>
Intercept	2.15	1.13	1.30	1.21	3.69***	1.03	2.49***	0.59	3.01***	0.66	3.31***	0.56
Continuity	0.68*	0.27	0.54	0.29	0.48	0.25	0.36*	0.14	0.20	0.16	-0.04	0.13
Emotional Event	-0.09	0.34	0.22	0.22	0.13	0.19	0.04	0.11	-0.10	0.12	-0.10	0.10
Continuity × Emotional Event	-0.44*	0.19	-0.28	0.20	-0.23	0.17	-0.13	0.10	0.08	0.11	0.05	0.09

Table S9. Simple slopes analysis of the moderating effect of emotional event on the relationship between relational cues and emotional interaction experiences

IV	DV	Emotional Event	Simple slope	<i>SE</i>	<i>p</i>
Self-disclosure	Emotional Attachment	Extreme	0.10	0.04	0.022
		Moderate	-0.05	0.05	0.297
Self-disclosure	Human-AI trust	Extreme	0.07	0.04	0.081
		Moderate	-0.05	0.04	0.237
Empathy	pleasure	Extreme	0.70	0.11	< 0.001
		Moderate	0.21	0.12	0.085
Empathy	Arousal	Extreme	0.57	0.12	< 0.001
		Moderate	0.16	0.13	0.213
Empathy	Dominance	Extreme	0.58	0.10	< 0.001
		Moderate	0.19	0.11	0.086
Empathy	Emotional attachment	Extreme	0.56	0.06	< 0.001
		Moderate	0.26	0.06	< 0.001
Empathy	Usability	Extreme	0.51	0.07	< 0.001
		Moderate	0.30	0.07	< 0.001
Empathy	Human-AI trust	Extreme	0.33	0.06	< 0.001
		Moderate	0.17	0.06	0.007
Humor	Pleasure	Extreme	0.24	0.09	0.007
		Moderate	-0.05	0.10	0.654
Meta-talk	Pleasure	Extreme	0.15	0.10	0.152
		Moderate	-0.18	0.10	0.093
Meta-talk	Emotional Attachment	Extreme	0.16	0.05	0.002
		Moderate	0.00	0.05	0.967
Continuity	Pleasure	Extreme	0.23	0.12	0.057
		Moderate	-0.21	0.17	0.204

Note. IV is the independent variable and DV is the dependent variable

4. Basic information of participants in Study 2

Table S10. Basic information of participants in structured interviews

Code	Gender	Age	Education	Occupation	Chatbots commonly used for emotional interaction
R01	Male	22	Undergraduate	Student	DeepSeek
R02	Male	21	Associate Degree	Student	Doubao
R03	Female	21	Undergraduate	Student	ChatGPT
R04	Female	21	Undergraduate	Student	ERNIE Bot
R05	Female	22	Undergraduate	Student	ERNIE Bot
R06	Female	22	Undergraduate	Student	Doubao
R07	Female	25	Undergraduate	Civil servants	Doubao
R08	Male	27	Undergraduate	Private Enterprises	Doubao
R09	Male	23	Undergraduate	Student	ChatGPT
R10	Female	21	Undergraduate	Student	Doubao
R11	Female	24	Undergraduate	Private Enterprises	Doubao
R12	Male	23	Undergraduate	Student	ChatGPT
R13	Female	23	Undergraduate	Student	ChatGPT
R14	Female	21	Undergraduate	Student	ChatGPT
R15	Male	23	Undergraduate	Student	Kimi
R16	Male	22	Master	Student	Doubao
R17	Male	22	Master	Public Institutions	Doubao
R18	Male	29	Master	Public Institutions	Doubao
R19	Female	20	Undergraduate	Student	BagelBell
R20	Male	23	Undergraduate	Student	DeepSeek

R21	Male	29	Undergraduate	private enterprises	Doubao
R22	Male	20	Undergraduate	Public Institutions	Doubao
R23	Male	37	Undergraduate	Private enterprises	Tencent Yuanbao
R24	Male	21	Undergraduate	Public Institutions	Doubao
R25	Male	20	Undergraduate	Student	Doubao

5. Online survey question

Table S11. Online Survey Questions

Category	Question
Relational cue	(1) <i>During emotional interactions, do the chatbot's human-like emotional and social behaviors improve your mood, making it more comfortable and pleasant? If you were to rate the impact of the chatbot's relational cues on your emotional interaction experiences on a scale of 1 to 10, what score would you give and why?</i> (2) <i>What is your view on the human-like emotional and social behaviors displayed by the chatbot?</i>
Self-disclosure	(1) <i>During emotional interactions, when the chatbot shares personal information (such as its functions, background, or personality), does your mood improve? If you were to rate the impact of the chatbot's self-disclosure on your emotional interaction experiences on a scale of 1 to 10, what score would you give and why?</i> (2) <i>What is your view on the chatbot's self-disclosure?</i>
Empathy	(1) <i>During emotional interactions, if the chatbot can fully understand and empathize with your emotional state (for example, by expressing understanding or responding to your feelings), does your emotional interaction experiences improve? If you were to rate the impact of the chatbot's ability to empathize on your emotional interaction experiences on a scale of 1 to 10, what score would you give and why?</i> (2) <i>What is your view on the chatbot's ability to empathize?</i>

- Humor (1) *During emotional interactions, do you feel that your mood improves when the chatbot makes jokes? If you were to rate the chatbot's effect on improving your emotional interaction experiences on a scale of 1-10, what score would you give and why?*
- (2) *What is your view on the chatbot's sense of humor?*
- (Dis)continuity (1) *During emotional interactions, do you feel frustrated or bothered if the chatbot suddenly changes the topic or fails to follow up on your statement? On a scale of 1 to 10, what score would you give the impact of the chatbot's discontinuous conversation on your emotional interaction experiences and why?*
- (2) *What is your view on the continuity of the chatbot's conversation?*
- Meta-talk (1) *During emotional interactions, does the chatbot improve your mood when it engages in self-reflection or comments on the progress of the conversation? If you were to rate the impact of the chatbot's meta-talk on your emotional interaction experiences on a scale of 1 to 10, what score would you give and why?*
- (2) *What is your view on the chatbot's meta-talk?*
-

6 Results of Online Survey of Study 2

Table S12. The impact of chatbot relational cues on emotional interaction experiences

Code	Reference	Definition	Example
Increase the Sense of Realism	14(56%)	Making the interaction feel more human-like.	<i>I feel like, at this moment, it's not just AI, but more like a companion always by my side (R07); It feels similar to conversing with a friend (R09); I would regard this software as a valuable companion in my life (R10).</i>
More Effectively fulfill Inner Emotional Needs	10(40%)	Effectively addressing users' emotional needs, fulfilling their desires for expression and sharing.	<i>This behavior makes the interaction very comfortable. For example, he cares about my feelings like a friend, and he's genuinely happy for me when I share happy things. This kind of interaction makes me feel fulfilled (R17).</i>
Foster Intimacy	8(32%)	Deepening and personalizing the emotional connection between users and the chatbot.	<i>It enables me to engage more closely and delve deeper into these issues (R04); I can resonate more easily with a chatbot that seems more human-like, without creating a barrier (R12).</i>
Enhance the Conversational Experience	6(24%)	Improving the fluency and engagement in human-AI interaction.	<i>Because during the process, I become more and more engaged in the conversation (R03); This would significantly enhance the experience of human-AI interaction (R15).</i>
Alleviate stress	3(12%)	Effectively reducing users' stress and worries, loosening conversational constraints.	<i>I can chat with it without feeling burdened, and life becomes more interesting, plus it doesn't invade your thoughts (R03).</i>

Feel Valued or Attended to	1(4%)	Feeling as though the chatbot genuinely cares for and values them	<i>It feels as if a person, not AI, is recognizing me (R12).</i>
Difficult to express	1(4%)	Making it difficult to express one's feelings.	<i>If human emotions and behaviors were involved, I might find it difficult to express my feelings (R23).</i>

Table S13. The impact of chatbot self-disclosure on emotional interaction experiences

Code	Reference	Definition	Example
More Comfort and Pleasure	14(56%)	Making the interaction more comfortable and pleasant	<i>During emotional communication, the chatbot's self-disclosure makes me feel comfortable and pleased (R03); If I could chat with such a chatbot, I would feel very happy (R14).</i>
Foster Intimacy	11(44%)	Perceiving psychological closeness and enhancing empathy experiences	<i>Because it makes me feel understood (R01); It makes me feel like we are getting closer (R02); It promotes intimacy and emotional attachment (R15).</i>
Increase the Sense of Realism	10(40%)	Making the chatbot appear more human-like, leading users to feel they are talking to another person rather than AI.	<i>I feel a more real sensation, like I'm chatting with another person (R12); It makes me feel like I'm really chatting with a friend. It really feels like a person, with its own personality and background (R14).</i>
Enhance Naturalness	4(16%)	Making the chatbot feel less rigid and awkward compared to traditional chatbots, and promoting bidirectional interaction.	<i>It doesn't make me feel awkward (R05); It makes it less stiff compared to other chatbots (R11).</i>
Reduced Psychological Stress	4(16%)	Reducing the psychological pressure in the communication environment, thus allowing for freer self- expression.	<i>During its self-disclosure process, I didn't feel mental pressure (R03); I can tell it things I can't say to others without worry (R10).</i>
Increase the Sense of Security	3(12%)	Feeling stable and protected	<i>I feel quite secure (R05).</i>
Improve Credibility of Suggestions	2(8%)	Increasing the value of the chatbot's suggestions	<i>It may increase the credibility of its suggestions in some matters (R13).</i>
Add fun	2(8%)	Breaking the monotony of fixed scenarios and activities, making conversations more interesting.	<i>Chatbot would be more interesting if it could self-disclose (R01); Chatbots can bring a sense of novelty by self-disclosing new scenarios and activities (R19).</i>

Still a Chatbot Program	2(8%)	Highlighting the gap between the expectation of the chatbot being human and the reality that it is AI.	<i>When I realize it's just a chatbot, there's a slight psychological gap (R04); Its self-disclosure is still just a program (R12).</i>
Increase communication burden	1(4%)	The information provided by the chatbot itself may create a psychological burden during the conversation.	<i>I don't want to know any details about it. Communication is easier when neither party knows the other (R23).</i>

Table S14. The impact of chatbot empathy on emotional interaction experiences

Code	Reference	Definition	Example
Improve Mood	21(84%)	Making users feel more comfortable and pleasant	<i>I feel it can improve my mood... I think my mood will get better (R05); It really did that... my mood... improved... very comfortable and pleasant (R09).</i>
Promote understanding	17(68%)	Enhancing users' perception of its understanding ability	<i>It can understand some of the things I want to express from my inner world, which makes me feel perfect (R07); It looks at things from my perspective (R11).</i>
Increase the sense of realism	11(44%)	Making the interaction feel more like talking to a human, increasing the sense of realism in the conversation	<i>I feel like it's more like a person, not a cold robot (R04); It feels more like chatting with a real person (R07).</i>
Provide Appropriate Advice	7(28%)	Providing more valuable and satisfying advice or guidance after understanding the user's emotions	<i>It would give me more useful advice (R03); It can fully understand me and offer suggestions on what should be done (R11); When it fully understands me and provides appropriate help, I feel more comfortable, pleasant, and satisfied (R15).</i>
Offer Comfort	7(28%)	Offering care and comfort more effectively when the user is feeling down	<i>It can understand me, feel my emotions, and comfort me (R10); It was very good at offering timely comfort (R21).</i>
Provide a Sense of Companionship	6(24%)	Providing a sense of companionship similar to that of a friend	<i>It fills the gap when my friends aren't around, providing companionship (R12).</i>
Better Listening and Response	9(36%)	Making it a better listener, allowing users to share things they may not be able to express to others in real life	<i>There are many things I can't share with people in my life but can share with AI (R10); I don't have to deal with humans anymore; I can have emotional exchanges with the chatbot (R07).</i>

Feel valued	3(12%)	Feeling as though the chatbot genuinely cares for and values them	<i>When I was feeling down, it accurately understood my grievances and gave me a warm response, making me feel valued and listened to (R22).</i>
Actively guide	3(12%)	Encouraging users to do the right thing and promote growth	<i>It understood my self-blame and encouraged me to grow from my mistakes (R24).</i>

Table S15. The impact of chatbot humor on emotional interaction experiences

Code	Reference	Definition	Example
Improve Mood	16(64%)	More comfort and pleasure	<i>When it tells jokes, I feel my mood improves (R04).</i>
Alleviate Stress	9(36%)	Effectively alleviating user' stress and worries, reducing conversational constraints	<i>After a tiring day, when I get home and see its jokes, all my troubles and stress suddenly vanish (R07); There are fewer constraints in the conversation (R08); It helps users relieve stress very well (R10).</i>
Increase the Sense of Realism	5(20%)	Making the interaction feel more like talking to a human	<i>Its sense of humor makes me feel like talking to them is like having a normal conversation with a human (R01); It feels more human-like (R04).</i>
Increase Intimacy, Trust, and Closeness	5(20%)	Effectively enhancing human-machine relationships	<i>If the chatbot can respond and participate in jokes, it would enhance my intimacy and trust in it (R15); I would also want to be friends with it and become dependent on it (R03).</i>
Add fun	4(16%)	Making conversations more interesting.	<i>Its humor makes emotional interaction fun (R17).</i>
Offer Comfort	3(12%)	Effectively providing comfort	<i>When it tells jokes, I feel a bit comforted and happy (R06).</i>
Transmission of Humor	1(4%)	Triggering similar joyful reactions in others	<i>I think smiles are contagious, and humor is the same. A joke that makes someone laugh heartily will improve their mood (R07).</i>
Consideration of Users' Emotional State	8(32%)	The effect of humor may vary depending on the user's emotional state	<i>When I'm feeling really sad, the jokes it tells make me want to laugh (R04); It can improve my mood when I'm feeling disappointed (R05); When it tells jokes in certain emotional states, I just can't laugh (R12).</i>
Consideration of Humor Application	6(24%)	Understanding users and appropriately and actively using humor in emotional communication can improve mood	<i>If it can appropriately use humor, it will make the communication atmosphere more relaxed and pleasant. but it should also control the use of humor (R15); If it knows how to cheer me up, it can improve my mood (R11); If it takes the initiative to joke, it can improve my mood (R13).</i>
Limited Humor Effect	4(16%)	Humor has a limited effect, not very significant, but still useful	<i>The content of its jokes doesn't make me laugh, but the way it tells jokes is always quite amusing (R10); It tells some dry jokes, the kind that are standard (R14).</i>

Table S16. The impact of chatbot meta-talk on emotional interaction experiences

Code	Reference	Definition	Example
Improve Dialogue Quality	10(40%)	Better recognition and timely correction of communication flaws to enhance the quality of conversation and improve communication effectiveness	<i>It makes me think about where my words went wrong (R04); If it can timely improve its shortcomings from the previous conversation, I think the effect is quite good (R13).</i>
Increase the sense of realism	9(36%)	Making the conversation more real and smooth, like interacting with a real person	<i>It feels a bit like a person, able to self-reflect (R05); Real humans also self-reflect when they make mistakes (R06); The dialogue felt less stiff and more natural (R20).</i>
Increase the depth of interaction	4(16%)	Fostering deeper understanding and meaningful exchange in the interaction.	<i>This reflective dialogue makes the conversation more profound (R22).A deep sense of awareness adds depth to the interaction (R24);</i>
Consideration of Users' Emotional State	3(12%)	If the user is irritated or upset, the chatbot's meta-talk may worsen the user's discomfort	<i>If I urgently want an answer but it says this, I feel a little disappointed (R04).</i>
No Significant Impact	2(8%)	The chatbot's meta-talk has no significant impact on the user's emotions	<i>I didn't feel anything special when it said this (R03).</i>
Ability to Respond	2(8%)	Even if it doesn't fully understand the user, at least it responds	<i>When I say something it doesn't understand, at least it doesn't leave me without a response (R06).</i>
Too Mechanical	4(16%)	Meta-talk feels unnatural and makes communication awkward	<i>It doesn't need to answer so mechanically (R10); If this behavior is not natural, it will make the conversation stiff (R15).</i>
Wear Down Patience	1(4%)	Frequent misunderstandings may wear down the user's patience	<i>If it keeps saying it doesn't understand, I'd get discouraged if I were patient enough (R08).</i>

Table S17. The impact of chatbot discontinuity on emotional interaction experience

Code	Reference	Definition	Example
Induce Negative Emotions	12(48%)	Discontinuous conversations lead to negative emotions, such as disappointment	<i>The conversation was going well, but when it suddenly changes the topic, I feel uncomfortable (R07).</i>
Affect Conversation Experience	12(48%)	Topic incoherence affects the flow of conversation and user engagement; continuous dialogue makes communication smoother	<i>I feel confused and lose the motivation to continue the conversation (R13); When this happens with the chatbot, it can disrupt the flow and atmosphere, making me feel the conversation isn't smooth (R15); If it maintains continuity, at least the communication will be smoother (R08). If it can maintain continuity, the conversation will feel more like talking to a real person (R03); Maintaining continuity feels like consulting a long-term, permanent psychologist or companion (R12).</i>
Affect the sense of realism	7(28%)	Maintaining conversation continuity makes the exchange feel more natural, like talking to a real person	<i>If it suddenly ends the topic when I don't feel like it's finished, I feel really disturbed (R06).</i>
Feel Disturbed	6(24%)	Sudden topic changes cause frustration or dissatisfaction	<i>If it suddenly ends the topic when I don't feel like it's finished, I feel really disturbed (R06).</i>
Feel Ignored/Valued	5(20%)	Changing the topic makes the user feel neglected or ignored, while maintaining continuity makes the user feel valued	<i>I feel like it didn't pay attention to what I said (R03); If it keeps the continuity, I feel like it's really listening to me (R10).</i>
Not Understanding the User	5(20%)	Changing the topic is seen as a failure to fully understand the previous conversation	<i>I feel like maybe it didn't really understand what we were talking about before (R14).</i>
No Significant Impact	5(20%)	The impact of discontinuity is minimal and acceptable.	<i>It's just a robot, still learning, so I won't complain just because it suddenly changed the topic (R18); If the topic is suddenly changed, I don't think it will have a big impact (R25).</i>
Shift Conversation Focus	2(8%)	Changing the topic is seen as an attempt to redirect the user's attention or shift the conversational angle	<i>When it changes the topic, it means it thinks the current issue has reached a critical point and can't continue to provide help, actually redirecting my attention (R12); I think I can change the angle to talk with it (R08).</i>

Note: To help participants more intuitively experience conversational continuity, the survey questions were designed to inquire about their experiences with discontinuous conversations.